

Assignment 10

Let V be a vectorspace over $\mathbb{C}$ with a pseudometric inner product $\langle \cdot, \cdot \rangle$

Let $M \subseteq V$ be nonempty and consider the orthogonal complement

$$M^\perp = \{v \in V : \langle v, u \rangle = 0 \forall u \in M\}$$

- a) Show that $M^\perp$ is a linear subspace and $M^\perp$ is closed with respect to the pseudometric corresponding to $\langle \cdot, \cdot \rangle$

First we show that it is a linear subspace:

Has 0 in it

since $\langle 0, u \rangle = 0 \forall u \in V$ we especially have $\langle 0, u \rangle = 0 \forall u \in M$ thus $0 \in M^\perp$

stable under addition and scalar multiplication

let $v, z \in M^\perp$ and $\beta \in \mathbb{C}$ then we have

$$\langle v + \beta z, u \rangle = \langle v, u \rangle + \beta \langle z, u \rangle = 0 \forall u \in M$$

(the first equality is since an inner product is linear in the first argument)

But that is the exact condition of being in $M^\perp$ and thus $v + \beta z \in M^\perp$

Thus $M^\perp$ is a linear subspace

Now we prove that $M^\perp$ is closed

We know that

$$M^\perp \text{ is closed} \Leftrightarrow (\{x_n\}_{n \in \mathbb{N}} \subseteq M^\perp \text{ and } x_n \text{ converges} \Rightarrow \lim x_n \in M^\perp)$$

So we simply let $\{x_n\}_{n \in \mathbb{N}} \subseteq M^\perp$ be a convergent series with limit $\lim x_n$

Thus we must have the following for every $n \in \mathbb{N}$

$$\langle x_n, u \rangle = 0 \forall u \in M$$

But lemma 9.1.9 gives us that

$$\langle \lim_{n \rightarrow \infty} x_n, u \rangle = \lim_{n \rightarrow \infty} \langle x_n, u \rangle = \lim_{n \rightarrow \infty} 0 = 0 \forall u \in M$$

And thus $x \in M^\perp$

Implying that $M^\perp$ is closed

- b) Assume that $\langle \cdot, \cdot \rangle$ is a (proper) inner product

Show then that

$$M^\perp = (\text{span}(M))^\perp = (\overline{\text{span}(M)})^\perp$$

By their definitions it is obvious that

$$M \subseteq \text{span}(M) \subseteq \overline{\text{span}}(M)$$

Giving the reverse structure when looking at the orthogonally complement since

$$\begin{aligned} M^\perp &= \{v \in V : \langle v, u \rangle = 0 \forall u \in M\} \supseteq (\text{span}(M))^\perp = \{v \in V : \langle v, u \rangle = 0 \forall u \in \text{span}(M)\} \\ &\supseteq (\overline{\text{span}}(M))^\perp = \{v \in V : \langle v, u \rangle = 0 \forall u \in \overline{\text{span}}(M)\} \end{aligned}$$

I will now prove

$$M^\perp \subseteq (\text{span}(M))^\perp$$

let $x \in M^\perp$ thus $\langle x, u \rangle = 0 \forall u \in M$

let $y \in \text{span}(M)$

since $y \in \text{span}(M)$ where $u_i \in M$

$$y = \sum_{i=1}^n \alpha_i \cdot u_i$$

then by (almost)linearity of the inner product (i dont remember the name of this attribute...)

$$\langle x, y \rangle = \langle x, \sum_{i=1}^n \alpha_i \cdot u_i \rangle = \sum_{i=1}^n \bar{\alpha}_i \cdot \langle x, u_i \rangle = \sum_{i=1}^n \bar{\alpha}_i \cdot 0 = 0$$

Implies $x \in (\text{span}(M))^\perp$

Thus $M^\perp \subseteq (\text{span}(M))^\perp$

$$(\overline{\text{span}}(M))^\perp \supseteq (\text{span}(M))^\perp$$

Let $x \in (\text{span}(M))^\perp$

let $y \in \overline{\text{span}}(M)$ and construct $\{y_n\}_{n \in \mathbb{N}} \subseteq \text{span}(M)$ such that $\lim_{n \rightarrow \infty} \|y - y_n\| = 0$

or $\lim_{n \rightarrow \infty} y_n = y$

for any n we have

$$\langle x, y_n \rangle = 0$$

Since $x \in (\text{span}(M))^\perp, y_n \in \text{span}(M)$

By 9.1.9 we then have

$$\langle x, y \rangle = \lim_{n \rightarrow \infty} \langle x, y_n \rangle = 0$$

And thus

$$x \in (\overline{\text{span}}(M))^\perp$$

Implies

$$(\overline{\text{span}}(M))^\perp \supseteq (\text{span}(M))^\perp$$

And we are done